

Statistical Learning

Theoretical Foundations and Model Validation

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BIostatistics RESEARCH UNIT



Part I

Foundations

- Statistical learning
- Prediction vs inference
- ERM
- Loss functions
- Bias–variance
- GLMs
- Bayesian learning

Part II

Model Validation

- Data splitting
- Cross-validation
- Bootstrap
- Nested CV
- ROC and AUC
- Calibration
- Decision curves

Part III

Regularization

- Ridge
- LASSO
- Elastic Net
- Feature selection
- Hyperparameter tuning
- Penalized likelihood
- Clinical example

The workshop progresses from theoretical foundations, through model validation, to modern methods for controlling complexity and improving prediction.

By the end of this session participants should be able to:

- Explain the statistical learning framework.
- Distinguish prediction from inference.
- Describe common loss functions.
- Explain empirical risk minimization.
- Explain the bias–variance tradeoff, including underfitting and overfitting.
- Understand regularization techniques.
- Recognize GLMs and Bayesian models within machine learning.
- Evaluate models using cross-validation and out-of-sample methods.

Part I

Foundations of Statistical Learning

Why Statistical Learning?

Modern biomedical research generates large and complex datasets.

Examples include

- HIV cohorts
- Genomics and transcriptomics
- Electronic health records
- Imaging data
- Longitudinal registries

Goal:

$$Y = f(X) + \varepsilon$$

Learn a function relating predictors to outcomes.

Machine learning is fundamentally statistical learning.

$$Y = f(X) + \varepsilon$$

where

- Y = outcome
- X = predictors
- $f(\cdot)$ = unknown relationship
- ε = random noise

Examples:

- Mortality prediction
- TB treatment outcomes
- Cancer recurrence

Inference

- Explain mechanisms
- Estimate effects
- Odds ratios
- Hazard ratios

Prediction

- Forecast outcomes
- Minimize error
- Accuracy prioritized
- Interpretability secondary

Smoking and lung cancer:

Inference asks:

"Does smoking increase risk?"

Prediction asks:

"Will this patient develop lung cancer?"

Learning is an optimization problem

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n L(Y_i, f(X_i))$$

Components:

- Loss function L
- Model class $\mathcal{F}$
- Average prediction error

Most machine learning algorithms differ only through

- ① the loss function;
- ② the function class.

$$\hat{f} = \arg \min \frac{1}{n} \sum L(Y_i, f(X_i))$$

- Y_i : observed outcome
- X_i : patient characteristics
- $f(X_i)$: predicted value
- L : penalty for prediction error
- $\mathcal{F}$: model family

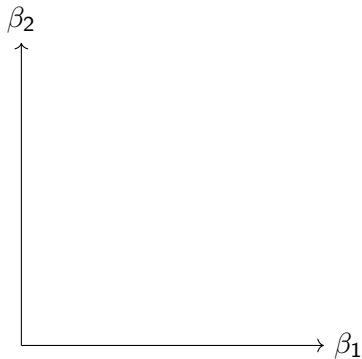
Example:

Predict 1-year mortality among HIV/TB patients using

- CD4 count
- Viral load
- BMI
- ART adherence

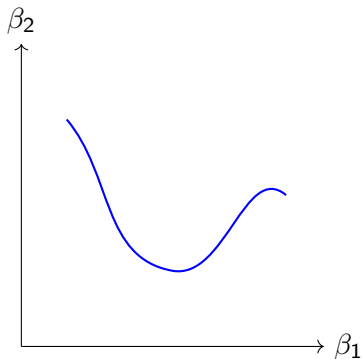
ERM: Geometric Interpretation

Training corresponds to searching for the minimum of a loss surface.



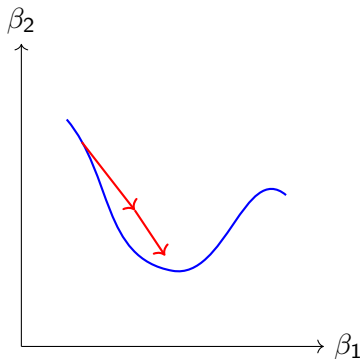
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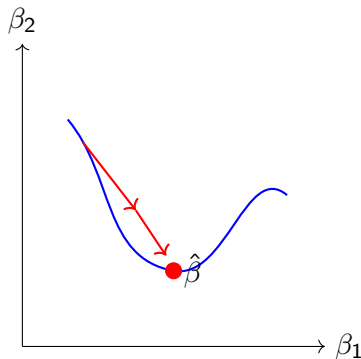
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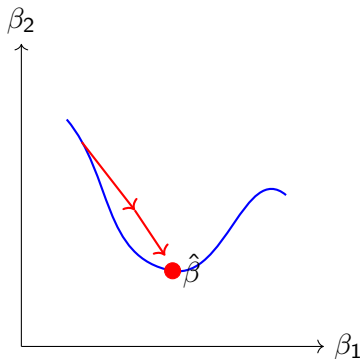
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Training corresponds to searching for the minimum of a loss surface.



ERM: Geometric Interpretation

Training corresponds to searching for the minimum of a loss surface.



Model fitting is equivalent to navigating a high-dimensional loss surface.

Statistical estimation and machine learning are closely related.

$$\hat{\theta}_{MLE} = \arg \max L(\theta)$$

Equivalent to

$$\hat{\theta} = \arg \min [-\log L(\theta)]$$

Examples:

- Logistic regression $\leftrightarrow$ cross-entropy loss
- Poisson regression $\leftrightarrow$ Poisson deviance
- Cox model $\leftrightarrow$ partial likelihood

ICU mortality model

- $AUC = 0.91$
- Excellent discrimination
- Poor calibration
- Risk probabilities too extreme

Clinical consequence:

Incorrect treatment allocation.

High accuracy does not guarantee clinical usefulness.
Calibration matters.

Why Model Validation is Necessary

ERM minimizes training error, but our goal is generalization.

- Training performance is optimistic
- Real-world data differs from training data
- Overfitting is inevitable in flexible models

Validation estimates how a model will perform on unseen data.

Loss Functions

The Engine Behind Learning

Why Do Loss Functions Matter?

Recall that empirical risk minimization seeks

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n L(Y_i, f(X_i)).$$

The loss function defines what constitutes an error.

Examples:

- Continuous outcomes $\rightarrow$ MSE
- Binary outcomes $\rightarrow$ Cross-entropy
- Count outcomes $\rightarrow$ Poisson deviance
- Survival outcomes $\rightarrow$ Partial likelihood

Changing the loss function changes what the model learns.

Regression models often minimize

$$L(Y, \hat{Y}) = (Y - \hat{Y})^2$$

which yields

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

Properties:

- Smooth and differentiable
- Penalizes large errors heavily
- Sensitive to outliers

Predicting CD4 counts from viral load and age.
Large prediction errors receive much larger penalties.

Alternative loss:

$$L(Y, \hat{Y}) = |Y - \hat{Y}|$$

leading to

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|.$$

Advantages:

- Robust to outliers
- Less influenced by extreme observations
- Median regression naturally arises

Hospital costs are highly skewed.
MAE is often preferred over MSE.

For binary outcomes, prediction error is measured by

$$L(Y, p) = -Y \log(p) - (1 - Y) \log(1 - p),$$

where

$$p = P(Y = 1|X).$$

Properties:

- Penalizes incorrect predictions heavily
- Produces probabilities between 0 and 1
- Equivalent to maximizing the logistic likelihood
- Widely used in classification

Cross-entropy measures how far predicted probabilities are from the true outcomes.

For binary outcomes,

$$L(Y, p) = -Y \log(p) - (1 - Y) \log(1 - p)$$

where

$$p = P(Y = 1|X)$$

Used in:

- Logistic regression
- Neural networks
- Deep learning

Predicting HIV mortality

$$Y = \begin{cases} 1, & \text{Death} \\ 0, & \text{Alive} \end{cases}$$

Estimate

$$p = P(\text{Death} | X)$$

using CD4 count, viral load, BMI
and ART adherence: X .

Count outcomes

Poisson models use the Poisson deviance.

Examples:

- Number of clinic visits
- Incidence counts
- Disease notifications

Time-to-event outcomes

Cox regression maximizes the partial likelihood

$$L(\beta) = \prod_i \frac{\exp(X_i\beta)}{\sum_{j \in R_i} \exp(X_j\beta)}.$$

Applications:

- Cancer survival
- HIV mortality
- Time to treatment failure

Huber Loss: A Compromise Between MSE and MAE

Huber loss combines quadratic and linear penalties.

Near zero:

$$L \approx (Y - \hat{Y})^2$$

Large errors:

$$L \approx |Y - \hat{Y}|$$

Advantages:

- Robust to outliers
- Stable optimization
- Frequently used in robust machine learning

Laboratory measurements may contain occasional errors.
Huber loss reduces their influence.

Comparison of Common Loss Functions

Outcome	Loss Function	Typical Models
Continuous	MSE	Linear regression
Continuous	MAE	Robust regression
Binary	Cross-entropy	Logistic regression
Counts	Poisson deviance	Poisson GLM
Survival	Partial likelihood	Cox PH model
Outliers	Huber loss	Robust ML

Loss functions determine the objective optimized during learning.

Bias–Variance Tradeoff

Understanding Overfitting and Generalization

Expected prediction error can be decomposed into

$$MSE = Bias^2 + Variance + \sigma^2.$$

where

- Bias measures systematic error;
- Variance measures sensitivity to data fluctuations;
- σ^2 represents irreducible noise.

Good prediction requires balancing bias and variance.

Simple models may fail to capture important patterns.

Characteristics:

- High bias
- Low variance
- Poor training performance
- Poor test performance

Using a straight line to describe a nonlinear relationship between age and blood pressure.

Underfitting = Model too simple

Complex models may fit random noise.

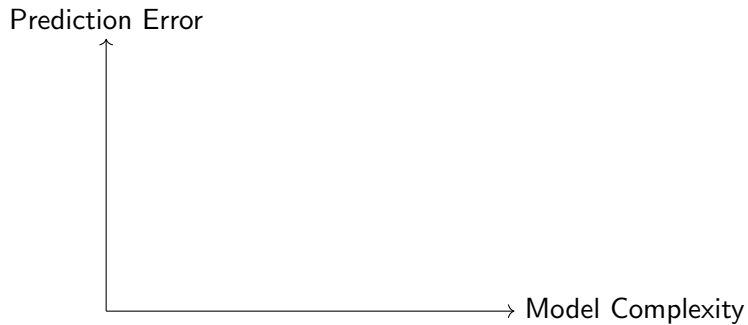
Characteristics:

- Low bias
- High variance
- Excellent training accuracy
- Poor external performance

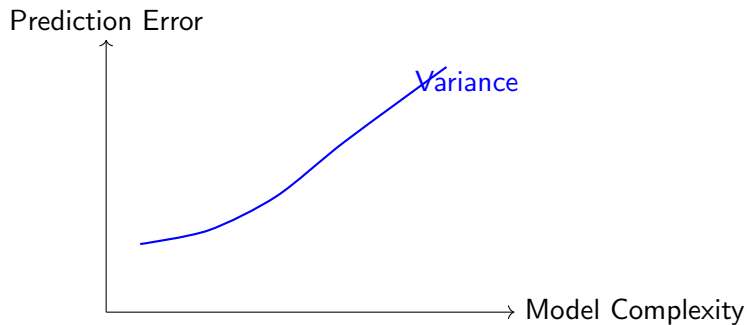
A prediction model developed on one hospital performs poorly elsewhere.

Overfitting = Model too complex

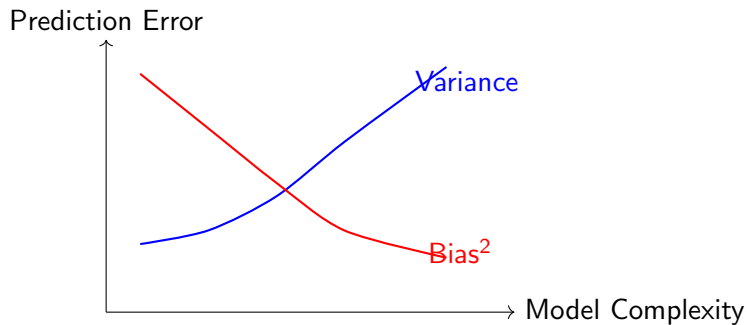
Bias–Variance Tradeoff



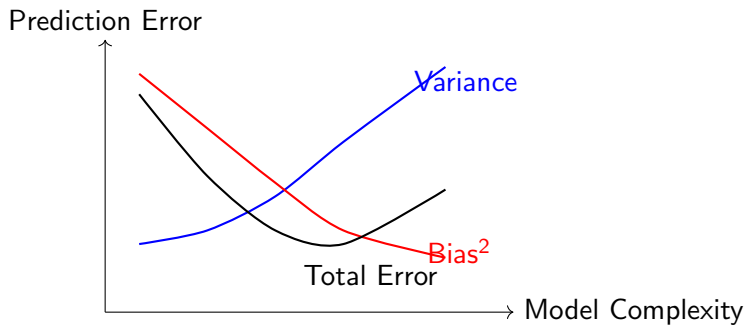
Bias–Variance Tradeoff



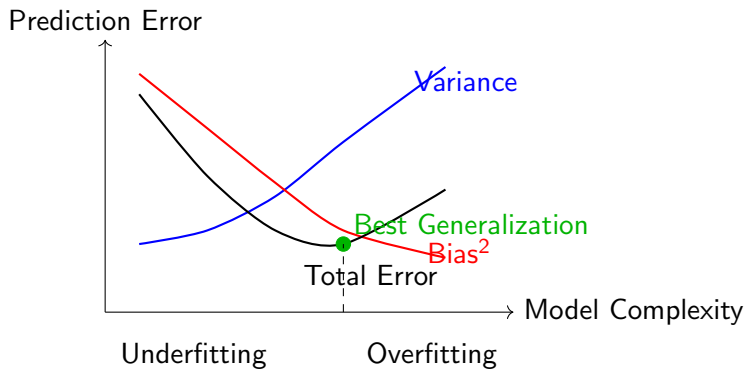
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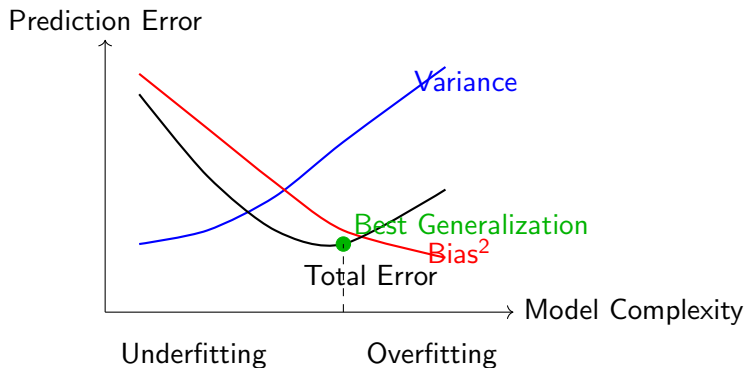
Bias–Variance Tradeoff



Bias–Variance Tradeoff



Bias–Variance Tradeoff



Optimal model complexity balances bias and variance and minimizes prediction error.

Clinical Example: Bias–Variance Tradeoff

Suppose we develop an ICU mortality model.

Model A

- Only age and sex
- High bias
- Underfits

Model B

- Hundreds of predictors
- Complex interactions
- High variance
- Overfits

Regularization seeks a balance between these two extremes.

Why move beyond linear regression?

Different outcomes require different probability models.

Outcome Type	Distribution	Model
Continuous	Gaussian	Linear regression
Binary	Bernoulli	Logistic regression
Counts	Poisson	Poisson regression
Overdispersed counts	Neg. Binomial	NB regression
Time-to-event	Hazard function	Cox model

GLM form:

$$g(\mu) = X\beta, \quad \mu = E(Y)$$

- $g(\cdot)$: link function
- $X\beta$: linear predictor

Binary outcomes in medicine

- HIV infection (Yes/No)
- Mortality (Alive/Dead)
- TB status (Positive/Negative)

Model:

$$\log \left(\frac{p}{1-p} \right) = X\beta$$

$$p = P(Y = 1|X)$$

Interpretation:

- coefficients = log-odds
- e^{β} = odds ratios
- ensures $0 \leq p \leq 1$

Count data in epidemiology

- HIV incidence counts
- Hospital admissions
- Disease events

Poisson model:

$$Y \sim \text{Poisson}(\lambda), \quad \log(\lambda) = X\beta$$

Issue: overdispersion ($\text{Var}(Y) > E(Y)$)

Solution:

- Negative Binomial regression
- adds dispersion parameter

Time-to-event outcomes

- Time to death
- Time to treatment failure
- Time to recurrence

Cox model:

$$h(t|X) = h_0(t) \exp(X\beta)$$

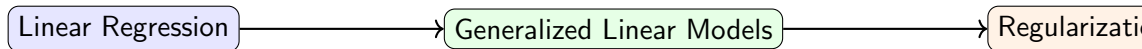
- $h_0(t)$: baseline hazard
- e^β : hazard ratio

Advantages:

- handles censoring
- semi-parametric
- widely used in clinical studies

From GLMs to Machine Learning

Most machine-learning methods extend ideas from GLMs.



Modern machine learning builds upon classical statistical modelling rather than replacing it.

Why Bayesian Statistics?

Frequentist inference treats parameters as fixed.

Bayesian inference treats parameters as random variables.

Bayes theorem:

$$P(\theta|Data) = \frac{P(Data|\theta)P(\theta)}{P(Data)}$$

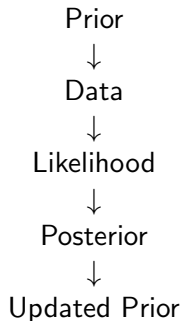
Combines:

- Prior knowledge
- Observed data
- Probability theory

Output:

Posterior Distribution

Bayesian learning is sequential.



Example:

- Prior belief about HIV prevalence
- Observe new survey data
- Update uncertainty
- Produce revised estimates

Credible Intervals versus Confidence Intervals

Frequentist

95% Confidence Interval

$(0.41, 0.60)$

Interpretation:

Repeated samples contain the true parameter 95% of the time.

Bayesian

95% Credible Interval

$(0.41, 0.60)$

Interpretation:

There is a 95% probability that the parameter lies inside the interval.

Bayesian intervals are often easier to interpret.

Posterior distributions rarely have closed forms.

Common algorithms:

- ① Markov Chain Monte Carlo (MCMC)
- ② Gibbs sampling
- ③ Metropolis-Hastings
- ④ Hamiltonian Monte Carlo
- ⑤ Variational Inference (VI)

Popular software:

- Stan
- JAGS
- BUGS
- PyMC

Bayesian Methods in Modern Machine Learning

Bayesian ideas underpin many AI methods.

Applications include:

- Bayesian neural networks
- Gaussian processes
- Ensemble learning
- Model averaging
- Uncertainty quantification
- Personalized medicine

Classical machine learning focuses on prediction.

Bayesian machine learning focuses on prediction together with uncertainty.

Part II

Model Validation

Assessing Generalizability and Performance

- ① Why model validation matters
- ② Internal validation
 - Train-validation-test split
 - Cross-validation
 - Bootstrap methods
- ③ External validation
- ④ Performance assessment
 - Confusion matrix
 - ROC curves and AUC
 - Precision-recall curves
 - Calibration
 - Decision curve analysis

A model that performs well on training data may fail on new data.

Common reasons:

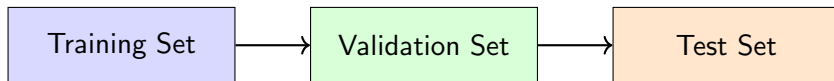
- Overfitting
- Small sample sizes
- Data leakage
- Distributional shifts
- Selection bias

Clinical consequence:

- Inflated performance estimates
- Poor reproducibility
- Unsafe clinical decisions

Validation assesses generalizability rather than goodness-of-fit.

Training, Validation and Test Sets



Typical split:

- Training: 60–70%
- Validation: 10–20%
- Testing: 20–30%

The test set should only be used once.

Cross-validation repeatedly evaluates the model.

Fold 1: Train Train Train Test

Fold 2: Train Train Test Train

Fold 3: Train Test Train Train

Fold 4: Test Train Train Train

Advantages:

- Efficient use of data
- Lower variance estimates
- Suitable for limited sample sizes

Common choice:

$$K = 5 \text{ or } 10$$

Bootstrap samples are generated with replacement.

Procedure:

- ① Sample observations with replacement
- ② Fit model
- ③ Evaluate performance
- ④ Repeat many times

Applications:

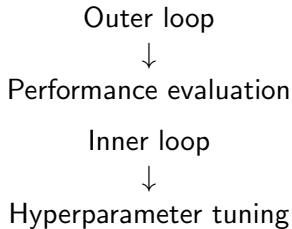
- Confidence intervals
- Optimism correction
- Internal validation
- Stability assessment

Popular approach:

Bootstrap .632

Nested CV separates:

- Model selection
- Model evaluation



Benefits:

- Avoids optimistic bias
- Provides unbiased estimates
- Recommended for machine learning studies

	Predicted +	Predicted -
Actual +	TP	FN
Actual -	FP	TN

Derived measures:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{Accuracy} = \frac{TP + TN}{N}$$

These metrics form the basis of diagnostic evaluation.

ROC curves summarize discrimination.

Sensitivity vs $1 - \textit{Specificity}$

Area under the curve:

$$0.5 \leq AUC \leq 1$$

Interpretation:

- 0.50 = random guessing
- 0.70 = acceptable
- 0.80 = good
- 0.90 = excellent

Widely used for prediction models.

PR curves are useful for rare outcomes.

Precision:

$$PPV = \frac{TP}{TP + FP}$$

Recall:

$$Sensitivity = \frac{TP}{TP + FN}$$

Examples:

- Cancer screening
- HIV incidence prediction
- Sepsis detection

PR curves are often preferable when classes are imbalanced.

Key idea: Discrimination is not enough.

Calibration asks:

Do predicted probabilities match observed outcomes?

Brier score:

$$BS = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{p}_i)^2$$

Lower values indicate better calibration.

Calibration plots compare:

- Predicted probabilities
- Observed event rates

A highly discriminative model can still be poorly calibrated.

Clinical utility extends beyond AUC.

Net benefit:

$$NB = \frac{TP}{N} - \frac{FP}{N} \left(\frac{p_t}{1 - p_t} \right)$$

where p_t is the risk threshold.

DCA compares:

- Treat all
- Treat none
- Prediction model

Application:

- Cancer risk prediction
- Cardiovascular disease
- HIV mortality models

Part III

Regularization and Feature Selection

Controlling Complexity and Preventing Overfitting

Complex models often fit random noise.

Consequences:

- Excellent training performance
- Poor external performance
- Unstable coefficient estimates
- Reduced reproducibility

Regularization adds a penalty to model complexity.

$$\text{Objective} = \text{Loss} + \lambda \times \text{Penalty}$$

Regularization improves generalization by shrinking parameters.

Ridge regression uses an L_2 penalty:

$$\hat{\beta} = \arg \min \left\{ RSS + \lambda \sum_{j=1}^p \beta_j^2 \right\}$$

Properties:

- Shrinks coefficients toward zero
- Retains all predictors
- Handles multicollinearity
- Stable when predictors are highly correlated

Applications:

- Genomics
- High-dimensional data
- Risk prediction

LASSO uses an L_1 penalty:

$$\hat{\beta} = \arg \min \left\{ RSS + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

Advantages:

- Automatic variable selection
- Sparse models
- Improved interpretability
- Useful when $p > n$

LASSO performs shrinkage and feature selection simultaneously.

Elastic Net combines Ridge and LASSO penalties.

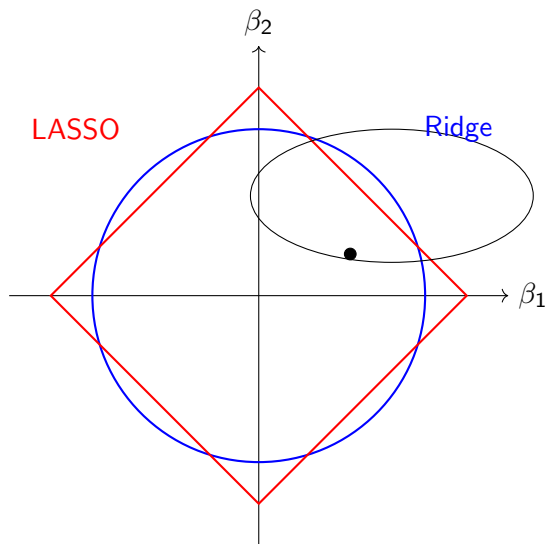
$$RSS + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$$

Advantages:

- Handles correlated predictors
- Performs variable selection
- More stable than LASSO
- Widely used in omics studies

Applications:

- Transcriptomics
- Proteomics
- Precision medicine



Goal:

Identify informative predictors while removing noise.

Methods:

- Stepwise selection
- LASSO
- Recursive feature elimination
- Random forest importance
- Stability selection

Benefits:

- Improved interpretability
- Reduced variance
- Faster computation
- Better prediction

Hyperparameters are chosen before training.

Examples:

- Ridge penalty λ
- Number of trees
- Learning rate
- Neural-network architecture

Common approaches:

- Grid search
- Random search
- Bayesian optimization

Nested cross-validation provides unbiased evaluation.

Regularization and Bias–Variance Tradeoff

Regularization increases bias slightly while reducing variance.

$$MSE = Bias^2 + Variance + \sigma^2$$

Effects:

- Slightly higher bias
- Much lower variance
- Better generalization
- Reduced overfitting

Good prediction is achieved through optimal complexity rather than maximal complexity.

Regularization can be viewed as penalized likelihood.

$$\ell_p(\beta) = \ell(\beta) - \lambda P(\beta)$$

Examples:

Method	Penalty
Ridge	$\sum \beta_j^2$
LASSO	$\sum \beta_j $
Elastic Net	Combination

Penalized likelihood unifies statistics and machine learning.

Clinical Example: Gene Expression Prediction

Objective: Predict breast cancer recurrence.

Data:

- 100 patients
- 10,000 genes
- Binary outcome

Challenge: $p \gg n$

Solution:

- LASSO logistic regression
- Variable selection
- Cross-validation
- External validation

Key insight:

Regularization stabilizes high-dimensional prediction.

Why it matters:

- avoids overfitting
- handles $p \gg n$
- improves generalization

From Regularization to Machine Learning Methods

Key idea: Modern ML extends regularization thinking.

- Regularization stabilizes linear models (LASSO, Ridge)
- ML methods extend this idea to complex function spaces

Core tradeoff revisited:

- Bias $\leftrightarrow$ variance
- Interpretability $\leftrightarrow$ flexibility
- Simplicity $\leftrightarrow$ predictive power

Why Decision Trees Need Stabilization

Problem: High variance

- Small data changes $\rightarrow$ very different trees
- Prone to overfitting

Interpretation:

- Low bias
- High variance

Motivation for ensembles:

Stabilize unstable learners by averaging or boosting.

- Ensemble of decision trees
- Uses bootstrapping + feature randomness
- Reduces variance via averaging

Key idea: variance reduction improves generalization.

- Sequential learning: each tree corrects errors
- Focuses on hard-to-predict cases
- Reduces bias incrementally

Key idea: boosting trades variance for reduced bias.

- Maximizes margin between classes
- Kernel trick enables nonlinear boundaries
- Strong theoretical guarantees

Key idea: controls complexity via margin maximization.

Model Evaluation in Clinical Machine Learning

Three key dimensions:

- **Discrimination:** ROC/AUC (ranking ability)
- **Calibration:** agreement with observed risk
- **Clinical utility:** decision curve analysis

Key message:

A good model must be *accurate, well-calibrated, and clinically useful*.

Statistical learning provides the theoretical foundation of modern machine learning.

- ① Prediction differs fundamentally from inference.
- ② ERM and likelihood theory underpin most learning algorithms.
- ③ Regularization controls overfitting and improves generalization.
- ④ Generalized linear models remain central to biomedical ML.
- ⑤ Bayesian methods quantify uncertainty.
- ⑥ Validation is essential for reproducibility.
- ⑦ Calibration and clinical utility matter as much as discrimination.

Machine learning should augment scientific reasoning rather than replace statistical thinking.

A word cloud featuring the phrase "Thank You" in numerous languages and colors. The central and largest text is "thank you" in red. Other prominent words include "danke" in blue, "gracias" in green, "teşekkür ederim" in pink, and "bedankt" in yellow. Smaller words in various colors include "ngiyabonga", "rahmat", "格雷斯", "welalin tack", "mersi", "barka", "vinaka", "spasibo", "blagodaram", "dank je", "misaotra", "malondo", "paldies grazi", "mahalo", "tapadh leat", "blagodarja", "asante", "manana", "tanks", "obrigado", "dziękuję", "sukriya", "kop khun krap", "go raibh maith agat", "dakujem", "eskerrik asko", "dankewol", "고맙습니다", "ti blagodarom", "ευχαριστώ", "shukriya", "dhanyavdagalu", "diolch", "tanemirt", "rahmet", "xiexie", "terima kasih", "arigatou", "aitäh", "sulpay", "dijere dieuf", "mochchakkeram", "mamnun", "trugarez", "dudu medidua", "sagolun", "najis tulke", "kam sah harnide", "salamat", "miersi", "dèkuj", "sobi", "dziękuję", "gracie", "hvala", "mauruuru", "maururu", "kiitos", "dankie", "dhanyavad", "nandi", "nandi", "baryarlala", "dzyakui", "faafetai lava", "misaotra", "malondo", "paldies grazi", "mahalo", "tapadh leat", "blagodarja", "asante", "manana", "tanks", "obrigado", "dziękuję", "sukriya", "kop khun krap", "go raibh maith agat", "dakujem", "eskerrik asko", "dankewol", "고맙습니다", "ti blagodarom", "ευχαριστώ", "shukriya", "dhanyavdagalu", "diolch", "tanemirt", "rahmet", "xiexie", "terima kasih", "arigatou", "aitäh", "sulpay", "dijere dieuf", "mochchakkeram", "mamnun", "trugarez", "dudu medidua", "sagolun", "najis tulke", "kam sah harnide", "salamat", "miersi", "dèkuj", "sobi", "dziękuję", "gracie", "hvala", "mauruuru", "maururu", "kiitos", "dankie", "dhanyavad", "nandi", "nandi", "baryarlala", "dzyakui", "faafetai lava".

Questions?

